Advanced IoT-Based Home Automation

Guide

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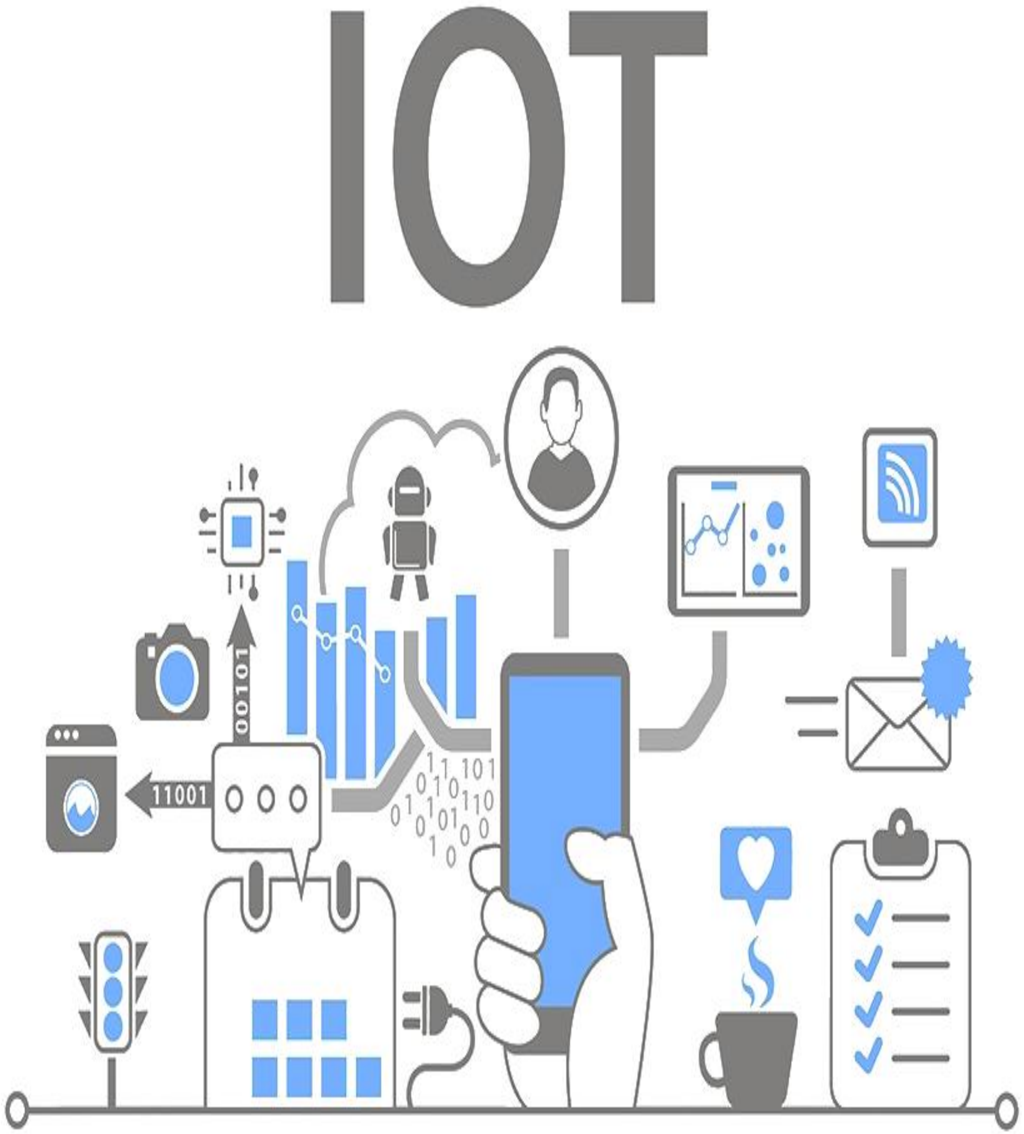
End of the presentation with some final thoughts

Introduction

- An network of physical objects—"things"—that are embedded with sensors, software, and other technologies for the purpose of connecting and exchanging data with other devices and systems over the internet.



Overview of Project

- "Advanced IoT Based Home Automation: Automatic Curtain System" is a project that uses IoT technologies to create a smart and automated curtain system. The system is based on the NodeMCU ESP8266, a WIFI-enabled microcontroller that controls a motor to open and close the curtains.
- The system can be controlled remotely using cloud-based services like Sinric Pro, which allows users to control their devices using voice commands or mobile apps. The project demonstrates the potential of IoT technologies to create innovative and convenient home automation solutions.

Literature Reviews

Understood IoT, Hardware, and software we need

Where we knew what is IoT and how it works! Also, know about some important hardware and software

Setup and Programming

Where we learned to setup and program our ESP8266 in Arduino IDE

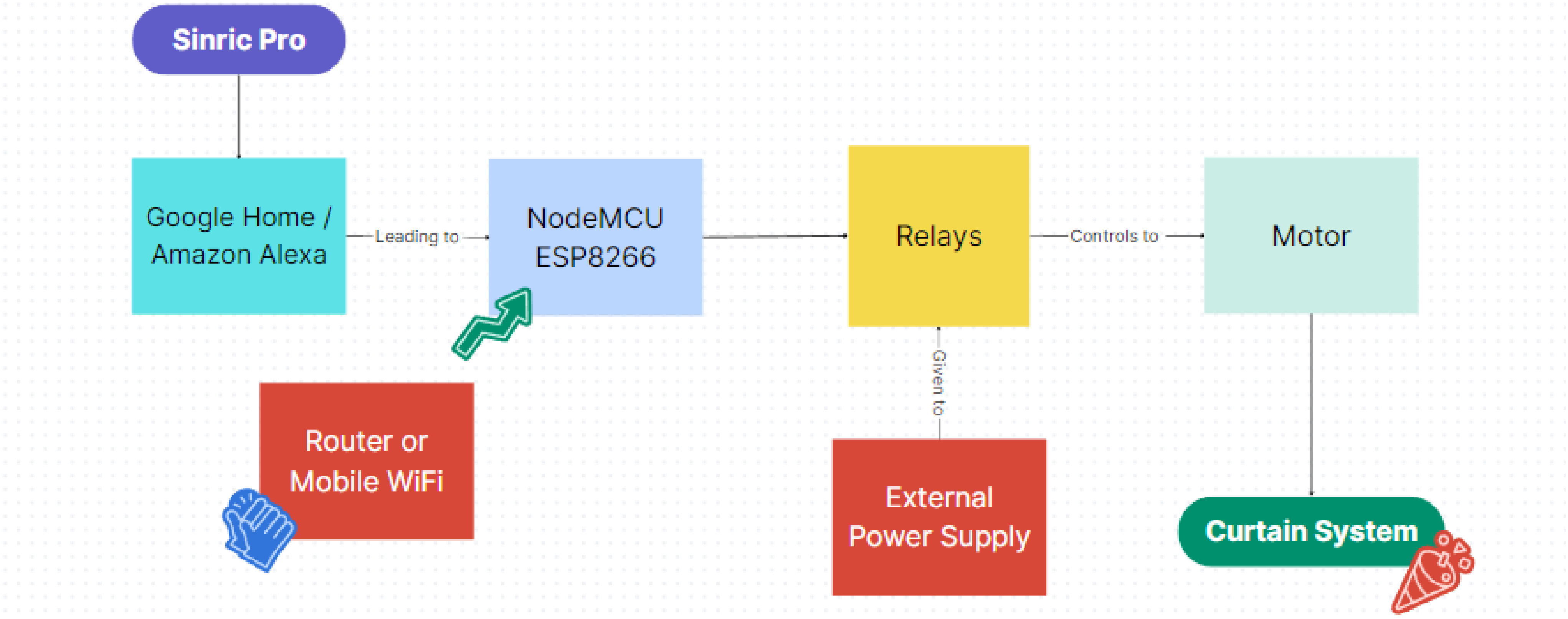
Know about Sinric Pro

Where we know about Sinric pro and how it works with different development boards

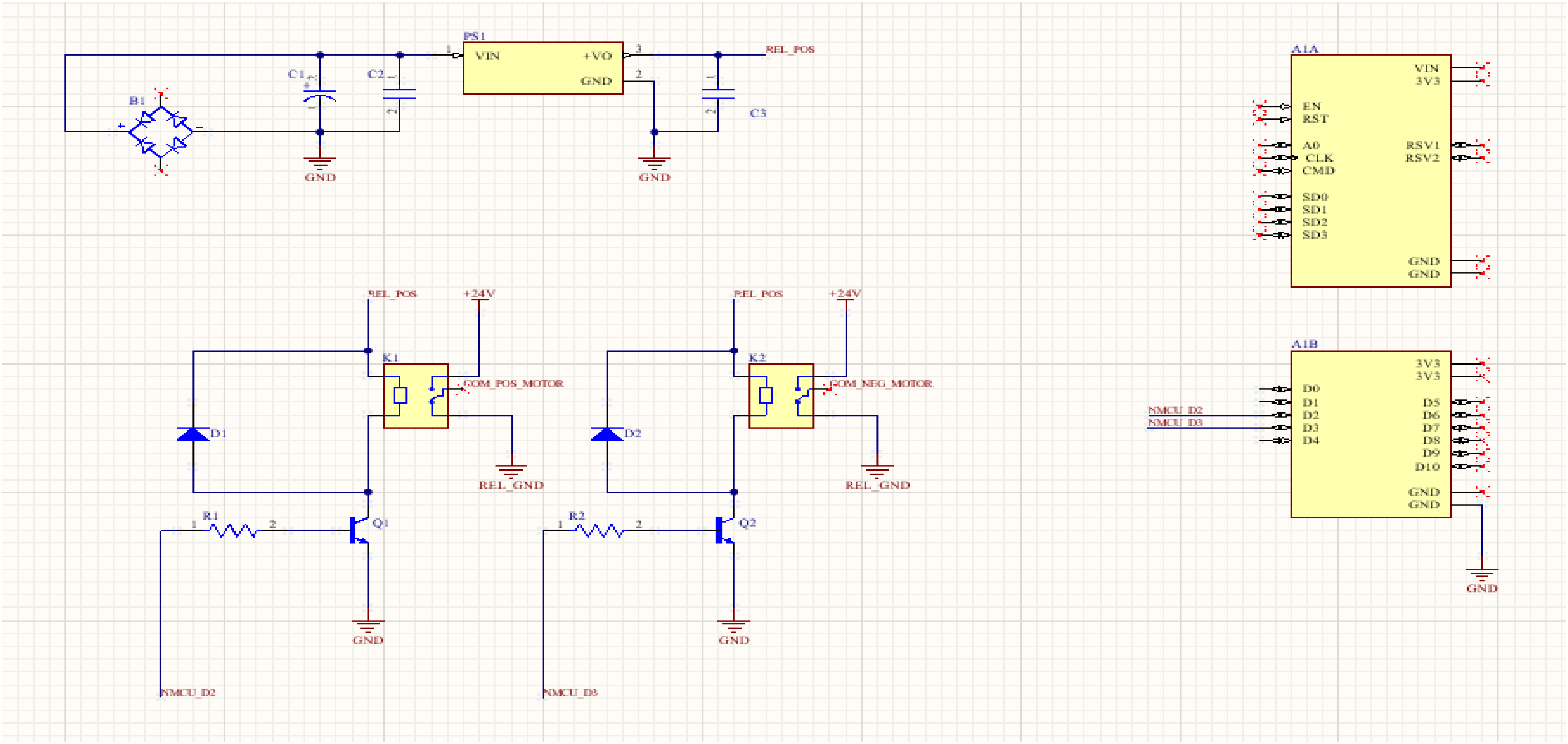
ESP8266 with Sinric Pro

To know how to work with Sinric Pro to control the NodeMCU ESP8266

The System Architecture



The Circuit Diagram



Required Electronics Components

(Hardware)

NodeMCU ESP8266

- ESP8266 is a low-cost wife microcontroller with a full TCP/IP stack. It has many GPIO (general purpose input output pins) for interfacing with different sensors.
- ESP8266 due to its low cost and good functionality is used a lot in prototyping IoT products.
- Its cross-functionality with Arduino makes it easy to program with Arduino IDE.

2-Channel Relay Module

- A relay is a simple electromechanical switch. While we use normal switches to manually close or open a circuit, a Relay is also a switch that connects or disconnects two circuits.
- But instead of a manual operation, a relay uses an electrical signal to control an electromagnet, which in turn connects or disconnects another circuit.

+24V DC Motor

- A DC motor is defined as a class of electrical motors that convert direct current electrical energy into mechanical energy.
- When electric fields and magnetic fields interact, a mechanical force arises. This is the principle on which the DC motors work.

Switched Mode Power Supply

- A switching power supply (SMPS) is an electronic circuit that converts power using a switching device that is turned on and off at a high frequency.
- A storage device such as an inductor or capacitor to provide power when the switching device is not available.

Required Software & Applications

(Software)

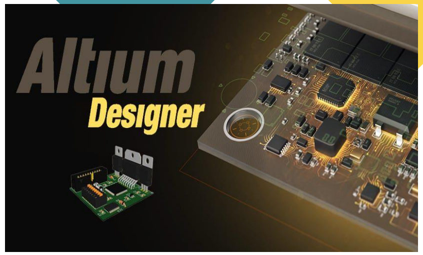
Required Software & Applications

Arduino IDE

For Programming / To program the NodeMCU ESP8266

Sinric Pro

Sinric Pro is a cloud-based service for IoT projects



Altium Designer

A PCB designing software to design the electronic PCBs

Required Software & Applications

Google Home

Application to control the NodeMCU ESP8266

Amazon Alexa

Application to control devices based using voice or web/app dashboards

Arduino IDE

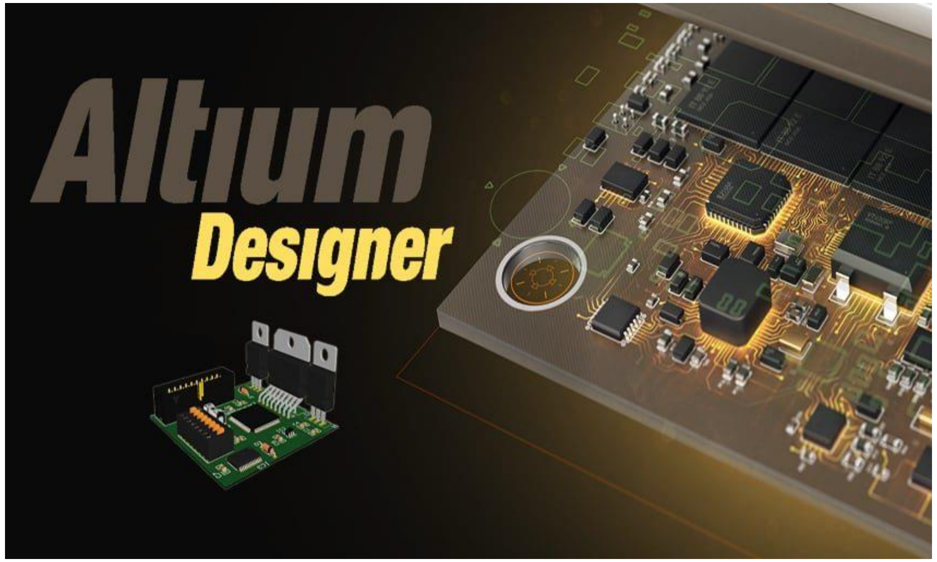
- The open-source Arduino Software (IDE) makes it easy to write code and upload it to the board. This software can be used with any Arduino board.
- Using Arduino IDE, we program our NodeMCU ESP8266.

Sinric Pro

- Sinric Pro is a software designed to allow easy integration of Amazon Alexa voice control into various devices and applications.
- Sinric Pro simplifies the process of integrating voice control into different software applications.

Altium Designer

- ❑ Altium is a software tool used for electronic circuit design and PCB (Printed Circuit Board) layout.
- ❑ It allows engineers and designers to create schematics, design custom components, and place and route PCBs in a single unified design environment.



Features

IoT-Based

WIFI Connectivity

Cloud-Based Control

Automatic Operation

Integration with Voice Assistance

Benefits

Increase efficiency and productivity in work

Save lot of time and efforts

Increased mobility and agility.

Drawbacks

Security Issues

Cost: Extremely expensive

Maintenance

Real World Applications

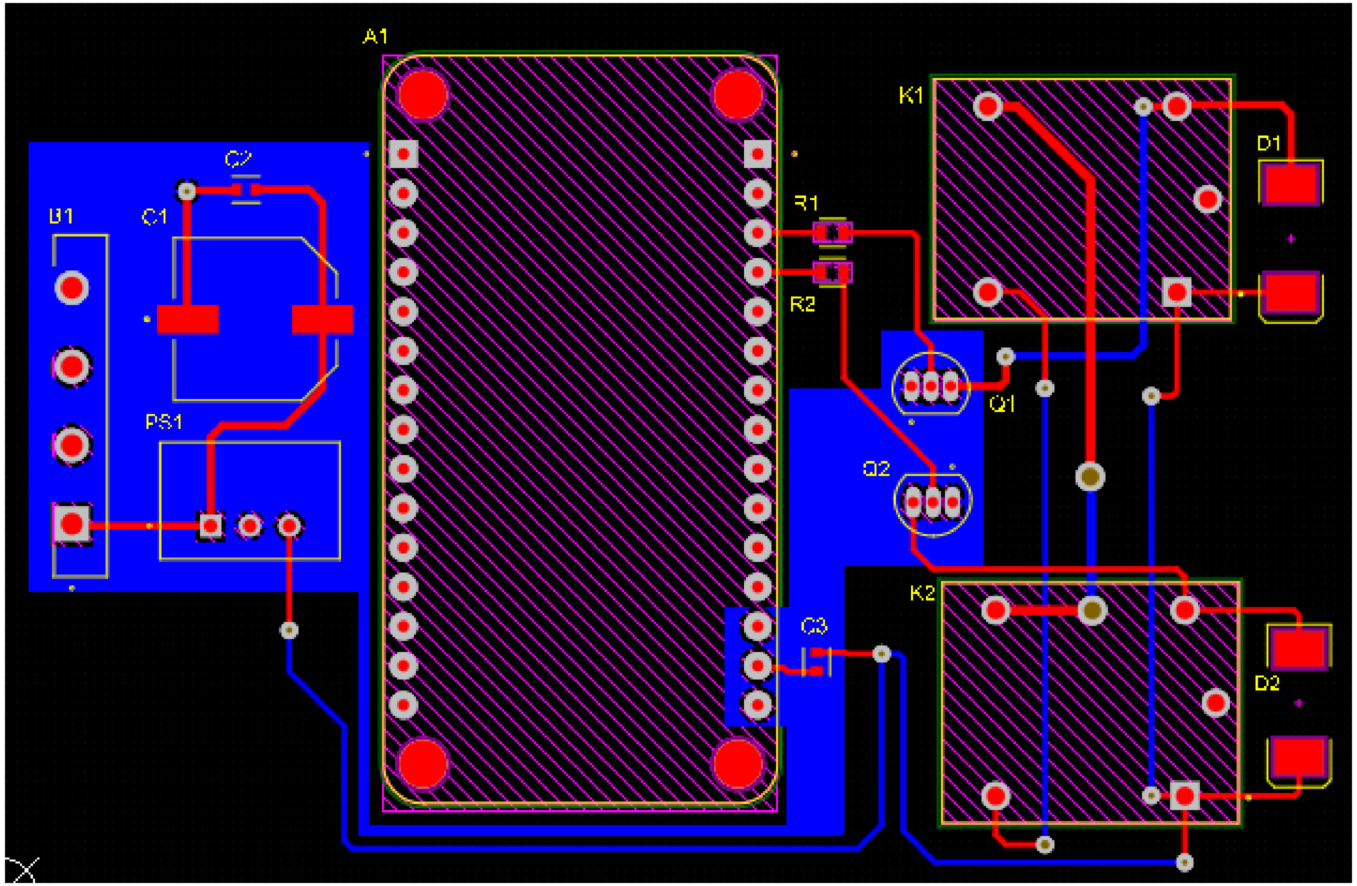
Lighting control

Household electronic
Appliances

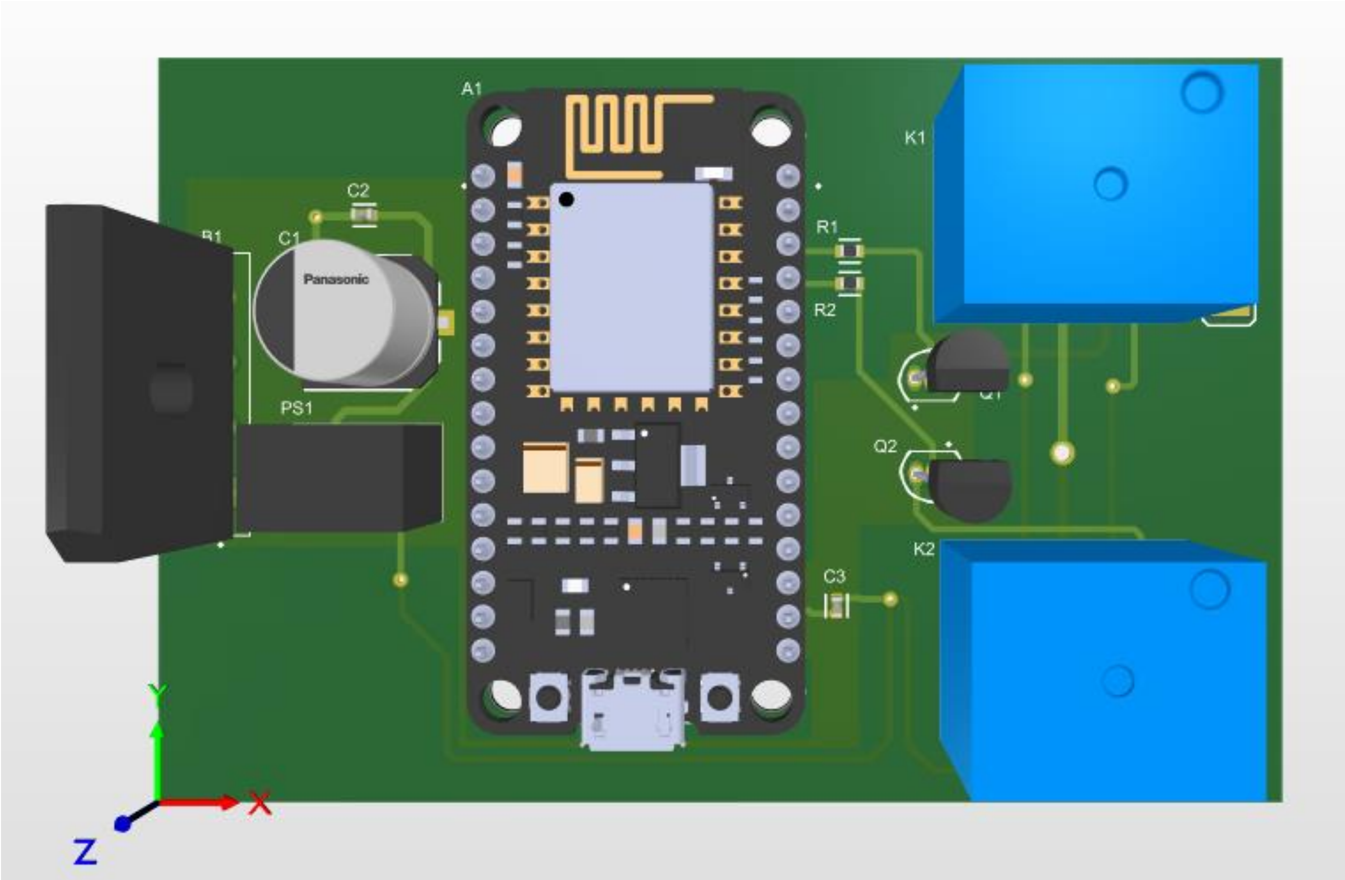
AI-driven digital
experiences

Let’s See The PCB Layout

2D View of the PCB



3D View of the PCB



References

[Learn about Sinric Pro]

[How to work with Sinric Pro to control ESP8266?]

[Programming]

[Sinric Pro with Google Home]

[Sinric Pro with Amazon Alexa]

Summary

✓ Introduction and Literature Reviews

We saw the basic introduction about home automation, IoT and about our project

✓ Required Hardware and Software

We discussed all the hardware parts and software that we will use to complete this project

✓ System Architecture and Circuit Diagram

We saw the simple block diagram of our IoT-Based System and also saw the circuit diagram

✓ Features, Advantages, Disadvantages and Real World Applications

Then we discussed the Features, advantages, disadvantages and application of the project

✓ PCB Layout

We have also seen the PCB design made with Altium Designer for this project

✓ References

Covered References!

Many thanks to you for giving your valuable time. We look forward to having you again in our next time. Your presence here definitely enriched this presentation.

And for this, we are grateful !